

Q10 Determine whether the linear transformation T is one-to-one, onto or neither.

(a) $T : \mathcal{P}_2 \rightarrow \mathbb{R}^3$ defined by $T(a + bx + cx^2) = \begin{bmatrix} 2a - b \\ a + b - 3c \\ c - a \end{bmatrix}$

Solution.

Consider $T(a + bx + cx^2) = \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}$, $\alpha, \beta, \gamma \in \mathbb{R}$.

Then

$$\begin{bmatrix} 2a - b \\ a + b - 3c \\ c - a \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} \implies \begin{bmatrix} 2 & -1 & 0 \\ 1 & 1 & -3 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}$$

$$\begin{aligned} \left[\begin{array}{ccc|c} 2 & -1 & 0 & \alpha \\ 1 & 1 & -3 & \beta \\ -1 & 0 & 1 & \gamma \end{array} \right] &\xrightarrow{R_2 \rightarrow R_2 - \frac{1}{2}R_1} \left[\begin{array}{ccc|c} 2 & -1 & 0 & \alpha \\ 0 & \frac{3}{2} & -3 & \beta - \frac{1}{2}\alpha \\ -1 & 0 & 1 & \gamma \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 + \frac{1}{2}R_1} \left[\begin{array}{ccc|c} 2 & -1 & 0 & \alpha \\ 0 & \frac{3}{2} & -3 & \beta - \frac{1}{2}\alpha \\ 0 & -\frac{1}{2} & 1 & \gamma + \frac{1}{2}\alpha \end{array} \right] \\ &\xrightarrow{R_3 \rightarrow R_3 + \frac{1}{3}R_2} \left[\begin{array}{ccc|c} 2 & -1 & 0 & \alpha \\ 0 & \frac{3}{2} & -3 & \beta - \frac{1}{2}\alpha \\ 0 & 0 & 0 & \gamma + \frac{1}{2}\alpha + \frac{1}{3}\beta - \frac{1}{6}\alpha \end{array} \right] = \left[\begin{array}{ccc|c} 2 & -1 & 0 & \alpha \\ 0 & \frac{3}{2} & -3 & \beta - \frac{1}{2}\alpha \\ 0 & 0 & 0 & \gamma + \frac{1}{3}\alpha + \frac{1}{3}\beta \end{array} \right] \end{aligned}$$

Thus, if $\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} \in T[\mathcal{P}_2]$, then $\gamma + \frac{1}{3}\alpha + \frac{1}{3}\beta = 0 \implies \gamma = -\frac{1}{3}\alpha - \frac{1}{3}\beta$.

Hence, $T[\mathcal{P}_2] = \left\{ \begin{bmatrix} \alpha \\ \beta \\ -\frac{1}{3}\alpha - \frac{1}{3}\beta \end{bmatrix} \mid \alpha, \beta \in \mathbb{R} \right\} = \left\{ \alpha \begin{bmatrix} 1 \\ 0 \\ -\frac{1}{3} \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ -\frac{1}{3} \end{bmatrix} \mid \alpha, \beta \in \mathbb{R} \right\} \neq \mathbb{R}^3$

since $\dim(T[\mathcal{P}_2]) = 2$ while $\dim(\mathbb{R}^3) = 3$.

Therefore, T is not onto.

Next, consider $T(a + bx + cx^2) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

From the rank-nullity theorem, we can see that $\dim(\ker(T)) = 3 - \dim(T[\mathcal{P}_2]) = 3 - 2 = 1$.

Hence, there are at least two distinct quadratic polynomials $p(x)$ and $q(x)$ s.t. $T(p(x)) = T(q(x)) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

Since $p(x) \neq q(x)$, T is not one-to-one.

Hence, T is neither one-to-one nor onto. □